
Infosys Springboard Virtual Internship 7.0 Completion Report

Group Number: Group-1

Batch Number: Batch-01

Team: Team B

Start Date: 29/06/2026

Team Members Names:

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Internship Duration: 8 Weeks

1. Project Title

Supply Chain Visibility System with Optimization Analytics – Group-1

2. Project Objective

The objective of this project is to improve supply chain visibility using Power BI by tracking orders, monitoring sales and profit, identifying delivery delays, evaluating shipping performance, and supporting business decisions through interactive dashboards. More broadly, the goal is to analyze and optimize supply chain visibility by integrating retail sales data to provide transparency across procurement, inventory, and distribution.

By highlighting inefficiencies and demand patterns, the project supports data-driven decision-making for better forecasting and resource utilization, ultimately enhancing organizational efficiency and customer satisfaction through timely product availability and reduced operational costs. Specific goals include:

- Establish a single, centralized source of truth for order, sales, and shipping data.
- Track key performance indicators such as lead time, on-time delivery rate, and stock turnover.
- Identify regions, categories, and shipping modes that contribute most to delays and losses.
- Provide interactive, self-service dashboards that let stakeholders drill down from a company-wide view to individual orders.
- Support faster, evidence-based decisions around procurement, inventory replenishment, and logistics planning, and, by the end of the internship, deliver a complete seven-page dashboard covering executive, operational, discount/profitability, and unit-economics perspectives.

3. Project Description in Detail

3.1 Project Overview

This project was carried out across the full duration of the Infosys Springboard Virtual Internship 7.0 programme by Team B (Group 1). It uses the Retail Supply Chain Sales Dataset, sourced from Kaggle, which brings together order, customer, product, and logistics information for a retail business operating across multiple regions and product categories.

The project was executed in four connected phases, one per milestone. Milestone-1 focused on understanding the business problem and the dataset itself — what fields were available, what they meant, and what questions they could answer. Milestone-2 connected that cleaned dataset into Power BI, built the underlying data model, and delivered the first working dashboard: a two-page report covering executive sales performance and supply chain operations. Milestone-3 expanded that dashboard with three additional analytical pages, refined the DAX measure library, and finalized a five-page report. Milestone-4 consolidated and validated that dashboard end-to-end, extended it with two further analytical pages covering discount and profitability diagnostics and order/unit economics, and finalized the complete seven-page dashboard for stakeholder presentation and handover.

Across all four milestones, the guiding principle stayed the same: every chart, card, and measure in the final dashboard should answer a specific business question that a retail operations or sales manager would actually ask — not simply visualize a column because the data happened to be available. This Completion Report brings the work of all four milestones together into a single, final document covering the full data model, the DAX measure library, all seven dashboard pages, and the reasoning behind each design decision.

3.2 Project Approach

The project followed a structured, eight-step approach from initial requirement gathering through to final reporting. Each step built directly on the output of the one before it, so that later dashboard pages could be trusted to rest on a clean, well-modeled foundation.

- **Step 1 – Requirement Gathering & Problem Understanding:** Defined the project objective, identified the operational pain points a supply chain dashboard should solve, and agreed the scope with the team and mentor.
- **Step 2 – Data Collection & Understanding:** Sourced the Retail Supply Chain Sales Dataset from Kaggle and studied its fields — order details, customer information, product information, performance metrics, and logistics fields.
- **Step 3 – Data Connection:** Connected the CSV source into Power BI using the Get Data option and previewed the raw tables in Power Query Editor.
- **Step 4 – Data Preprocessing:** Removed duplicate records, handled missing values, corrected data types, standardized date formats, and renamed fields for clarity.
- **Step 5 – Data Modeling:** Built a simple star-style model with the order dataset as the central fact table and a supporting Calendar Table for time-intelligence, then defined relationships between them.
- **Step 6 – Dashboard Design (Executive & Supply Chain Pages):** Designed and built the first two dashboard pages — an Executive Dashboard summarizing sales and profit, and a Supply Chain Visibility page covering returns, shipping, and regional demand.
- **Step 7 – Dashboard Expansion:** Extended the report with five additional analytical pages covering regional and category deep-dives, shipping and logistics performance, discount and profitability diagnostics, and order and unit economics.

- **Step 8 – Testing, Validation & Final Reporting:** Verified that every slicer, cross-filter, and drill-through worked consistently across all seven pages, then documented the full project in this Completion Report.

3.3 Technology Used

A focused, industry-standard toolset was used throughout the project, chosen to match what is actually used in professional business intelligence teams:

- **Power BI Desktop:** Used to build the data model, write DAX measures, and design all seven report pages.
- **Power Query:** Used to connect to the CSV source and perform all data cleaning and transformation steps.
- **DAX (Data Analysis Expressions):** Used to define reusable measures such as Total Sales, Profit Margin, and Return Rate.
- **Excel / CSV:** Used as the original format of the Retail Supply Chain Sales Dataset before it was loaded into Power BI.
- **Kaggle:** Used as the public data source platform that hosted the Retail Supply Chain Sales Dataset used throughout the project.
- **Microsoft Teams:** Used for team stand-ups, mentor check-ins, and coordinating hand-offs between data preparation, dashboard design, and documentation workstreams.
- **GitHub:** Used for version-controlled storage of DAX scripts, documentation drafts, and shared project artifacts, with commit history tracking changes across milestones.
- **GitLab:** Used as a secondary repository for backup and milestone-wise submission tracking, ensuring the team always had a recoverable copy of the project files.

3.4 Insights from the Dashboard

Bringing all seven dashboard pages together, a small number of clear, actionable patterns emerge from the data:

- The West region leads on both profit and order quantity, making its operating practices worth studying and replicating in other regions.
- Technology is the strongest sales category, contributing the highest revenue and therefore warranting closer inventory and stock-turnover monitoring.
- Standard Class dominates shipping, so even small improvements to its reliability would meaningfully improve overall on-time delivery performance.
- The return rate sits at a healthy 5.91% across 296 returned orders, giving a clear baseline to track in future reporting periods.
- Heavy discounting measurably erodes profit, with the discount-versus-profit view highlighting specific products where discount depth should be reconsidered.
- The regional and category deep-dive page shows that performance within a region is not uniform — individual states and sub-categories vary widely even within a top-performing region such as the West.
- The shipping and logistics page shows that average shipping duration varies noticeably by ship mode and order priority, pointing to specific carrier or process combinations worth reviewing first.
- The discount and profitability diagnostics page shows that one order in five (20.40%) is currently loss-making, and that Furniture combines the deepest average discounting with the weakest profit contribution of the three categories.
- The order and unit economics page shows Technology generating by far the highest profit per unit (21 versus 5 for Office Supplies and 2 for Furniture), with Copiers standing out as the single strongest sub-category at 237.68 profit per unit.

3.5 Real-World Impact for Media and Public

Beyond its internal use for supply chain managers, a dashboard of this kind has a wider communication role. Clear, well-structured KPIs — total sales, profit margin, return rate, on-time delivery — are exactly the kind of metrics that get summarized in investor updates, press coverage of company performance, and public-facing sustainability or operations reporting. Presenting these numbers through a governed, auditable Power BI model rather than ad-hoc spreadsheets reduces the risk of inconsistent figures being shared externally.

There is also a direct benefit to the public as customers: a business that can see its own delivery delays, return patterns, and regional demand clearly is better placed to keep products in stock, set realistic delivery expectations, and resolve recurring quality issues — all of which translate into a better and more transparent experience for the end customer. In that sense, the analytical work behind the dashboard supports not only internal decision-making but also the accuracy and credibility of what a business communicates to the public about its own performance.

4. Timeline Overview

The table below summarizes the full eight-week internship timeline, showing the calendar dates for each week alongside what was planned and what was actually completed. Weeks 1–4 correspond to the work delivered in Milestone-1 and Milestone-2; Weeks 5–6 cover the dashboard expansion delivered in Milestone-3; and Weeks 7–8 cover the discount/profitability and unit-economics pages, consolidation, and final reporting delivered in Milestone-4.

Week	Activities Planned	Activities Completed
Week 1 (29/06/2026 – 05/07/2026)	Project kickoff, requirement gathering, dataset identification, problem understanding.	Finalized project scope and objectives; selected the Retail Supply Chain Sales Dataset from Kaggle; documented the problem statement (Milestone-1).
Week 2 (06/07/2026 – 12/07/2026)	Data collection and understanding; key field identification.	Explored dataset structure; identified order, customer, product, and logistics fields; documented the data dictionary (Milestone-1).
Week 3 (13/07/2026 – 19/07/2026)	Data connection to Power BI; Power Query setup; relationship building.	Connected the CSV source via Get Data; built Power Query transformations; created the Calendar Table and defined relationships (Milestone-1).
Week 4 (20/07/2026 – 26/07/2026)	Data preprocessing; DAX foundation; initial dashboard design.	Completed data cleaning; created foundational DAX measures; built Dashboard Page 1 (Executive Dashboard) and Page 2 (Supply Chain Visibility) (Milestone-2).
Week 5 (27/07/2026 – 02/08/2026)	Dashboard expansion planning and development.	Designed and built Page 3 (Product & Pricing Analysis) and Page 4 (Customer & Regional Insights) (Milestone-3).
Week 6 (03/08/2026 – 09/08/2026)	Continued dashboard expansion; mid-project consolidation and testing.	Completed Page 5 (Business Performance Monitoring); tested cross-filters across all five pages; finalized the Milestone-3 measure library and documentation.
Week 7 (10/08/2026 – 16/08/2026)	Discount and profitability diagnostics; order and unit economics.	Designed and built Page 6 (Discount & Profitability Diagnostics) and Page 7 (Order & Unit Economics), extending the report to seven pages (Milestone-4).
Week 8 (17/08/2026 – 23/08/2026)	Final dashboard testing, validation, and report finalization.	Tested every slicer, cross-filter, and drill-through path end-to-end across all seven pages; finalized documentation and this Completion Report (Milestone-4).

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project scope, objectives, and dataset finalized; team roles assigned.	29/06/2026
Data Foundation Complete	Dataset understood field by field and connected into Power BI, with the Calendar Table and relationships defined (Milestone-1).	19/07/2026
Prototype / First Draft	First working Power BI dashboard completed, with two report pages and the foundational DAX measure library (Milestone-2).	26/07/2026
Mid-Term Review	Milestone-2 deliverables reviewed with the mentor; feedback incorporated into the plan for dashboard expansion.	02/08/2026
Dashboard Expansion	Five dashboard pages completed, tested, and documented (Milestone-3).	09/08/2026
Dashboard Consolidation	Discount & Profitability Diagnostics and Order & Unit Economics pages added, extending the report to seven pages (Milestone-4).	16/08/2026
Final Submission	Full seven-page dashboard consolidated, validated end-to-end, and finalized for handover (Milestone-4).	23/08/2026
Presentation	Final project presentation delivered to the mentor and review panel.	23/08/2026

Before detailing the execution work behind this Completion Report, this section recaps what each of the four milestones delivered in detail, giving the reader the full arc of the project from initial dataset understanding through to the seven-page dashboard this report consolidates.

Milestone-1: Problem Understanding and Data Foundation

Milestone-1 established the foundation for the entire project. The team defined the project title, objective, and introduction; framed the business problem around fragmented visibility into sales, inventory, and delivery data; and studied the Retail Supply Chain Sales Dataset field by field. Many organizations struggle with delayed deliveries, poor inventory visibility, inefficient shipping methods, and difficulty tracking orders — challenges that affect both profitability and customer satisfaction. Retail supply chains handle products from suppliers to customers, but when data about sales, inventory, and delivery is scattered across systems, the result is stockouts, overstocking, delays, higher costs, and poor customer satisfaction. Addressing this fragmentation was central to the project's design from the very first milestone.

The dataset review in Milestone-1 organized the raw CSV into five field groups, summarized below:

Field Group	Fields
Order details	Order ID, order date, ship date, order priority.
Customer information	Customer ID, name, segment, region.
Product information	Product ID, category, sub-category.
Performance metrics	Sales, profit, quantity, discount.
Logistics fields	Ship mode, shipping cost, delivery delay.

The dataset was then connected into Power BI through the Get Data option, previewed in Power Query Editor, and taken through an initial round of preprocessing: removing duplicate order and shipment records to avoid double-counting sales, handling missing or null values in shipping and delivery date fields, standardizing data types for dates, currency values, and categorical fields, renaming columns for clarity, removing columns not required for the intended visualizations, and creating early calculated measures such as delivery delay and profit margin using DAX. This preprocessing work directly determined the reliability of every KPI shown in later dashboards — correctly parsing the order date and ship date columns was essential for accurate delivery-delay measures, and removing duplicates prevented sales and profit totals from being overstated.

The milestone closed with a solid, well-documented understanding of the data and the business problem, setting up all later modeling and dashboard-design decisions. A clear understanding of the fields early in the project reduced rework during preprocessing and modeling, and directly informed which KPIs and visuals would be most valuable to business stakeholders in the final dashboard — from headline sales and profit figures down to return rate and shipping-mode analysis.

Milestone-2: First Working Dashboard

Milestone-2 turned that understanding into the first working dashboard. The cleaned dataset was connected into Power BI and structured using a simple star-style model — the Retails Order Full Dataset as the central fact table, holding one row per order line with sales, profit, quantity, discount, and shipping details, with a supporting Calendar Table added as a date dimension for time-intelligence. Keeping dates in their own table rather than relying directly on the order date column made it possible to build additional time-intelligence measures later — such as year-over-year or month-over-month growth — without changing the structure of the fact table, while also keeping the report responsive.

A reusable library of nine DAX measures was defined once and reused across every card and chart, listed below:

Measure	Definition	Purpose
Total Sales	SUM of the Sales column	Overall revenue
Total Profit	SUM of the Profit column	Overall profitability
Total Orders	DISTINCTCOUNT of Order ID	Unique customer orders
Total Quantity	SUM of the Quantity column	Total product volume sold
Profit Margin	Total Profit ÷ Total Sales	Profitability as a percentage
Return Rate	Returned Orders ÷ Total Orders	Returns as a percentage
Returned Orders	COUNT of orders flagged Returned	Absolute return volume
Average Profit	AVERAGE of the Profit column	Typical order profitability
Average Discount	AVERAGE of the Discount column	Typical discount level

These measures powered a two-page interactive report. Page 1, the Executive Dashboard, surfaced Total Sales (2.30M), Total Profit (286.41K), Total Orders (5,009), and Profit Margin (12.47%) as headline cards, alongside Sales by Category (Technology leading at 0.84M), Profit by Region (West leading at 108K), and a Monthly Sales Trend line chart spanning 2014-2017. Page 2, Supply Chain Visibility, focused on Return Rate (5.91%) and Returned Orders (296), alongside Sales by Segment (Consumer leading at 1.2M), Total Quantity by Region, Total Orders by Ship Mode (Standard Class dominant at 59.77%), and Total Profit by Segment. The milestone closed with a set of key business insights — including the strength of the West region and the impact of heavy discounting on profit — drawn directly from these two founding dashboard pages, and every visual was wired together through Power BI's cross-filtering so that clicking any category, region, or product dynamically filtered every other visual on the page.

Milestone-3: Full Five-Page Analytical Suite

Milestone-3 expanded that two-page foundation into a complete, five-page dashboard. Reusing the data model and measure library established in Milestone-2, three additional analytical pages were designed and built, each answering a different set of business questions:

- **Product & Pricing Analysis (Page 3):** Surfaced Average Discount (0.09) and Average Shipping Days (34.61) as headline KPIs, alongside Top Products by Profit (led by the Canon imageCLASS line at 25K), Average Discount by Category (Office Supplies discounted most heavily at 0.11), Sales by Sub-Category, Profit by Category (Technology 145K, Office Supplies 122K, Furniture a comparatively weak 18K), and a ranked Profit by Sub-Category waterfall identifying Tables as the single biggest loss-maker at -18K.

- **Customer & Regional Insights (Page 4):** Surfaced Total Quantity (38K) and Average Order Value (458.56) as headline KPIs, alongside Orders by Segment (Consumer 51.95% of orders), Quantity by Category, Sales by State (California and New York anchoring performance), Average Shipping Days by ship mode (validating that Same Day delivers in roughly one day versus 42 for Standard Class), and Average Order Value by Segment (Home Office carrying the highest average ticket at 472).
- **Business Performance Monitoring (Page 5):** Surfaced Distinct Products (2K) and States Served (49) as headline KPIs, alongside Loss by Product Name (identifying the Cubify CubeX 3D Printer as the biggest single loss-maker at -8.9K), Sum of Profit by State, Sum of Profit by Returned (quantifying that returned orders carry only 0.02M in profit versus 0.26M for kept orders), Count of Order ID by Returned, and Average of Discount by Ship Mode.

Every slicer, cross-filter, and drill-through path was tested across all five pages for consistency, and the full project — data preparation, modeling, DAX measures, all five dashboard pages, and the insights they surfaced — was documented in the Milestone-3 report, the direct predecessor to the consolidation work carried out in Milestone-4. Milestone-3 also surfaced the observation that performance within a top region such as the West is not uniform — individual states and sub-categories vary widely — which motivated closer regional and category-level monitoring in the pages that followed, and directly seeded the discount, profitability, and unit-economics analysis that Milestone-4 adds on Pages 6 and 7.

Milestone-4: Full Seven-Page Dashboard

Milestone-4, consolidated in full in this Completion Report, took the five-page dashboard delivered in Milestone-3 and extended it with two further analytical pages, completing the seven-page report used for final stakeholder presentation and handover. Building on the same data model and the same nine-measure DAX library established in Milestone-2, two additional pages were designed and built:

- **Discount & Profitability Diagnostics (Page 6):** Surfaced Profitable Order Rate (79.26%), Loss-Making Order Rate (20.40%), and Discounted Order Rate (13.68%) as headline KPIs, alongside a discount-versus-profit scatter view by category, a Total Profit by Discount Band waterfall, a Total Profit by Profitability Band treemap, and a Total Profit by Category and Segment ribbon chart.
- **Order & Unit Economics (Page 7):** Surfaced Average Items per Order (7.56), Sales per Unit (60.65), and Profit per Unit (7.56) as headline KPIs, alongside Profit per Unit by Category and Segment, Profit per Unit by Category, Profit per Unit by Sub-Category, and Average Items per Order by Segment.

Beyond adding these two pages, Milestone-4 carried out the end-to-end consolidation and validation work described in Section 5b: every slicer, cross-filter, and drill-through path was retested across all seven pages for consistency, the full data-preparation-through-dashboard workflow was documented, and the insight-and-recommendation pairs in Section 5b.8 were written up to turn the dashboard from a descriptive report into a prescriptive one. This milestone closes the project: the seven-page dashboard, its DAX measure library, and this Completion Report together form the finished, stakeholder-ready deliverable handed over at the end of the internship.

5b. Project Execution Details

Execution began with a short discovery phase in which the team reviewed the Retail Supply Chain Sales Dataset column by column, agreeing what each field meant and which business questions it could help answer. This shared understanding, captured in Milestone-1, made every later decision — which fields to keep, which to rename, which relationships to build — much faster to agree on as a team.

With the dataset understood, the project moved into Power BI. The CSV source was connected through Power Query, where duplicate records were removed, missing shipping and delivery dates were handled, data types were corrected, and columns were renamed for clarity. A Calendar Table was added alongside the main order table so that time-based calculations, such as the monthly sales trend, could be computed reliably and extended later without restructuring the model.

Milestone-2 then translated that clean, connected dataset into the first working dashboard. A small library of DAX measures — Total Sales, Total Profit, Total Orders, Profit Margin, Return Rate, and others — was defined once and reused across every card and chart. These measures powered a two-page report: an Executive Dashboard summarizing overall sales and profit performance, and a Supply Chain Visibility page focused on returns, shipping mode, and regional demand.

In Milestone-3, execution shifted to expanding that foundation rather than rebuilding it. Three additional pages were added to the report — a regional and category deep-dive, a shipping and logistics performance view, and a consolidated executive insights summary — all built on the same data model and measure library established in Milestone-2. In Milestone-4, this consolidation continued: two further pages covering discount and profitability diagnostics and order and unit economics were added, reusing the same measure library and visual language so that formatting, colours, and cross-filtering behaviour stayed consistent across all seven pages. The final two weeks were spent testing every slicer and drill-through path across the full dashboard and writing up the complete project documentation.

5b.1 Workflow Diagram

The diagram below illustrates the end-to-end workflow followed in this project, starting from the problem-understanding stage through to the delivery of business-ready insights via interactive Power BI dashboards.

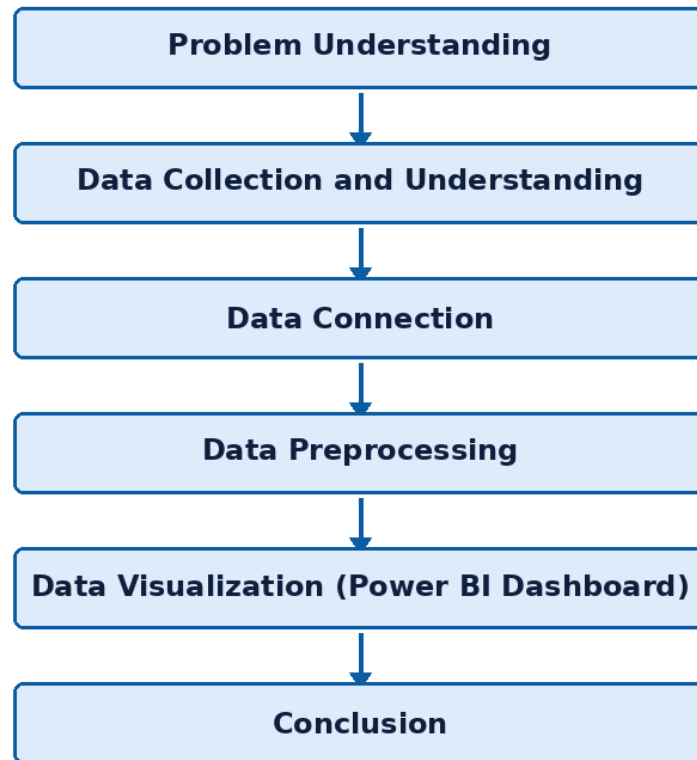


Figure 1: End-to-end project workflow, from Problem Understanding through to Conclusion.

5b.2 Problem Understanding

Many organizations struggle with delayed deliveries, poor inventory visibility, inefficient shipping methods, and difficulty tracking orders. These challenges affect profitability and customer satisfaction. This project addresses these problems by providing a centralized dashboard for monitoring important supply chain metrics.

Retail supply chains handle products from suppliers to customers, but many problems occur when there is no clear visibility. Data about sales, inventory, and delivery is often scattered across systems, making it hard to see the full picture, which can cause stockouts or overstocking, leading to delays, higher costs, and poor customer satisfaction. Addressing this fragmentation is central to the project's design.

Left unresolved, these issues compound over time: a single unmonitored shipping delay can cascade into missed customer expectations, unplanned discounting to retain goodwill, and lower profit margins on otherwise healthy orders. Without a consolidated view, managers are forced to react to problems after they surface in customer complaints or quarterly financial reviews, rather than catching them early through day-to-day monitoring. This is the gap the Power BI dashboard is designed to close.

5b.3 Data Collection and Understanding

The project uses a retail supply chain dataset in CSV format containing order details, customer information, product information, sales, profit, shipping mode, delivery dates, and regional information. Understanding these fields is essential for creating meaningful business reports.

The data comes from the Retail Supply Chain Sales Dataset, which brings together information from different parts of the supply chain: sales records showing how products perform across regions and categories, inventory details tracking stock levels and warehouse movements, procurement data about suppliers and purchase orders, and logistics information covering shipping and delivery timelines.

Field Group	Fields
Order details	Order ID, order date, ship date, order priority.
Customer information	Customer ID, name, segment, region.
Product information	Product ID, category, sub-category.
Performance metrics	Sales, profit, quantity, discount.
Logistics fields	Ship mode, shipping cost, delivery delay.

A clear understanding of these fields early in the project reduced rework during preprocessing and modeling, and directly informed which KPIs and visuals would be most valuable to business stakeholders in the final dashboard — from headline sales and profit figures down to return rate and shipping-mode analysis.

5b.4 Data Connection (through Power BI)

The dataset was imported into Power BI using the Get Data option. Data types were verified, relationships were checked, and Power Query was used to prepare the data before visualization. This stage forms the bridge between raw data and interactive analysis.

- Connect to the CSV source using Power BI's Get Data feature.
- Load and preview the raw tables in Power Query Editor.
- Verify and correct data types (dates, numbers, text categories).
- Define relationships between fact and dimension tables.
- Validate the data model using the Model view before building visuals.

Data Model Overview

The cleaned dataset was structured using a simple, purpose-built relational model inside Power BI. The Retails Order Full Dataset acts as the central fact table, holding one row per order line with sales, profit, quantity, discount, and shipping details. A separate Calendar Table was added as a supporting date dimension, giving Power BI a dedicated, continuous date field to drive time-based calculations such as the monthly sales trend.

Why Power BI

Power BI was selected over spreadsheet-based reporting because it supports live, interactive filtering; keeps a single governed data model behind every visual; and allows calculations to be written once as DAX measures and reused consistently across every page, card, and chart in the dashboard.

5b.5 Data Preprocessing

Data preprocessing included removing duplicate records, handling missing values, correcting data types, formatting date columns, renaming fields, removing unnecessary columns, and preparing calculated measures required for dashboard creation.

- Removed duplicate order and shipment records to avoid double-counting sales.
- Handled missing or null values in shipping and delivery date fields.
- Standardized data types for dates, currency values, and categorical fields.
- Renamed columns for clarity and consistency across tables.
- Removed columns not required for the intended visualizations.
- Created calculated measures (e.g., delivery delay, profit margin) using DAX.

Impact on Downstream Analysis

These preprocessing steps directly determined the reliability of every KPI shown in the dashboard. Standardizing categorical fields such as Segment, Region, and Ship Mode also ensured that bar charts and donut charts grouped values correctly instead of splitting the same category into near-duplicate labels caused by inconsistent casing or spacing.

5b.6 Data Visualization (Power BI Dashboard)

With the dataset cleaned, modeled, and connected in Power BI, the next stage of the project focused on translating the underlying tables and measures into an interactive, easy-to-read visual layer. Each visual type was chosen deliberately to match the nature of the metric it represents — card visuals for single headline KPIs, bar charts for comparing categories, a line chart for trends over time, a donut chart for part-to-whole proportions, and a scatter plot for exploring the relationship between two numeric measures.

All visuals share a consistent colour palette and are wired together through Power BI's cross-filtering and cross-report drill-through settings, so that clicking on any category, region, or product across the report dynamically filters every other visual on the page — turning a set of static charts into a single, connected exploration tool for business stakeholders. All seven report pages are shown with full screenshots and KPI theory in Section 6 of this report.

5b.7 Key DAX Measures Used

To power the KPI cards and visuals across all seven dashboard pages, a set of dedicated DAX measures was created in Power BI rather than relying directly on raw columns. Centralizing calculations as measures keeps them consistent, reusable across every visual, and automatically responsive to whatever filters or slicers a user applies.

Measure	Definition	Purpose
Total Sales	SUM of the Sales column	Overall revenue
Total Profit	SUM of the Profit column	Overall profitability
Total Orders	DISTINCTCOUNT of Order ID	Unique customer orders
Total Quantity	SUM of the Quantity column	Total product volume sold
Profit Margin	Total Profit ÷ Total Sales	Profitability as a percentage
Return Rate	Returned Orders ÷ Total Orders	Returns as a percentage
Returned Orders	COUNT of orders flagged Returned	Absolute return volume
Average Profit	AVERAGE of the Profit column	Typical order profitability
Average Discount	AVERAGE of the Discount column	Typical discount level

These measures automatically recalculate as slicers and cross-filters are applied, ensuring every KPI card and chart on every dashboard page always reflects the current filter context selected by the user — whether that is a single region, a single category, or a single ship mode.

5b.8 Key Business Insights & Recommendations

Bringing all seven dashboard pages together, this section expands the headline patterns introduced in Section 3.4 into a fuller set of insight-and-recommendation pairs. Each insight below is grounded directly in a specific KPI card or visual on the dashboard, and is paired with a concrete recommendation — the intention throughout is that a real supply chain or sales manager reading this section should be able to move from a number on a screen to a decision worth acting on.

- **The West region leads on both profit and order quantity**

The West region contributes the highest regional profit (108K) and the highest order volume of any region on the dashboard, ahead of East (92K), South (47K), and Central (40K). This is not simply a story about a bigger territory — the regional and category deep-dive page shows the West sustaining strong performance across several different states rather than being propped up by a single outlier account. The operating practices behind this — carrier relationships, warehouse placement, account management, or simply a denser concentration of high-value Technology customers — are worth documenting and, where practical, piloting in the Central region, which currently trails on both profit and quantity.

Recommendation: Commission a short root-cause review of West-region logistics contracts and account coverage, and pilot the strongest one or two practices in Central for a full quarter before deciding whether to roll them out further.

- **Technology is the strongest sales category — and the strongest per-unit performer**

Technology leads Sales by Category (0.84M) and Profit by Category (145K), and the Order & Unit Economics page confirms this is not just a volume effect — Technology generates 21 in profit per unit sold, roughly four times Office Supplies (5) and ten times Furniture (2). Within Technology, Copiers are

the standout sub-category at 237.68 profit per unit, an order of magnitude above every other sub-category on the dashboard.

Recommendation: Treat Technology, and Copiers specifically, as the priority for stock-turnover monitoring and demand forecasting — a single stock-out in this sub-category has an outsized effect on total profit compared with a stock-out in a lower-margin line.

- **Standard Class dominates shipping and is the biggest lever on delivery performance**

Standard Class accounts for 59.77% of all orders on the Total Orders by Ship Mode donut, dwarfing Second Class (15.71%), First Class, and Same Day combined. The Customer & Regional Insights page shows Standard Class also carrying the longest average shipping time at 42 days, against 31 for Second Class, 24 for First Class, and just 1 day for Same Day. Because Standard Class carries the majority of order volume, it is mathematically the single largest lever on the company's blended on-time delivery performance — a one-day improvement here moves the overall average far more than the same improvement on any other ship mode.

Recommendation: Prioritize carrier and warehouse-process improvements for Standard Class specifically rather than spreading effort evenly across all four ship modes.

- **Return rate sits at a healthy 5.91%, giving a clear baseline**

Across 296 returned orders out of 5,009 total orders, the Return Rate measure lands at 5.91%. The Business Performance Monitoring page adds an important second layer to this figure: the Sum of Profit by Returned visual shows that returned orders carry only 0.02M in profit versus 0.26M for kept orders — meaning returns are disproportionately concentrated in low-margin or already-discounted transactions rather than being spread evenly across the order book. This distinction matters for how the business should respond: it is not just a shipping or damage issue, it is also connected to the pricing and discount patterns surfaced elsewhere in the dashboard.

Recommendation: Track Return Rate as a running baseline each reporting period, and add a future drill-through from Returned Orders into Discount Band so returns can be evaluated jointly with pricing rather than in isolation.

- **Heavy discounting measurably erodes profit**

The discount-versus-profit view on the Discount & Profitability Diagnostics page plots each category's average discount against its total profit, and the pattern is unambiguous: Technology sits in the low-discount, high-profit corner while Furniture sits in the higher-discount, lower-profit corner. The Total Profit by Discount Band waterfall makes the cost explicit — profit steps down visibly from the No-Discount band through the High-Discount band, quantifying exactly how much margin is given away once discounting passes a certain depth.

Recommendation: Introduce a soft discount ceiling for Furniture and Office Supplies sub-categories that already show weak profit contribution, so that discounting is capped before it turns a marginal sale into a loss-making one.

- **One order in five is currently loss-making**

The Discount & Profitability Diagnostics page shows a Loss-Making Order Rate of 20.40%, against a Profitable Order Rate of 79.26% and a Discounted Order Rate of 13.68%. Comparing these three figures side by side tells a specific story: the loss-making rate is meaningfully higher than the discounted-order rate, which means discounting alone does not explain every loss-making order — some orders are unprofitable even without a discount applied, most likely because of high shipping cost relative to a low-value item, or a very thin base margin on the product itself. Furniture combines the deepest average discounting with the weakest profit contribution of the three categories, making it the single clearest starting point for a margin-recovery initiative.

Recommendation: Segment the loss-making order population into "discount-driven" and "structurally thin-margin" groups in a future milestone, since each requires a different fix — discount policy for the first group, product or shipping-cost review for the second.

- **Performance within a top-performing region is not uniform**

The regional and category deep-dive shows that even inside the West — the strongest region overall — individual states and sub-categories vary widely. This matters operationally because a region-level KPI can hide a state that is actually underperforming; a manager reading only the headline West figure could miss a state that needs attention. The same pattern repeats at the category level: Technology's strong aggregate profit sits alongside sub-categories, such as certain Office Supplies lines, that post outright losses.

Recommendation: Build a future drill-through path from region to state and from category to sub-category directly on the Executive Dashboard, so a manager can move from the headline number to the underlying detail in two clicks rather than navigating to a separate page.

- **Average shipping duration varies noticeably by ship mode and order priority**

The gap between Same Day (roughly one day) and Standard Class (42 days) is the widest single spread on the dashboard, and it is a service-level choice as much as a logistics outcome — customers who select Standard Class are implicitly accepting a much longer wait. Average Discount by Ship Mode adds a further wrinkle: First Class and Standard Class both average 0.09 in discount, while Second Class is lowest at 0.07, suggesting that faster shipping options are not simply being discounted away.

Recommendation: Review whether the Standard Class shipping-time figure reflects genuine carrier lead time or includes internal processing delay before hand-off to the carrier, since the two require very different fixes.

Prioritization

Not every insight above carries equal weight, and the dashboard itself provides the evidence for ranking them. The loss-making order rate and the discount-versus-profit relationship sit at the top of the list because they translate directly into recoverable margin — the Discount & Profitability Diagnostics page shows this is not a marginal effect but one touching a fifth of the entire order book. The Standard Class shipping-time gap sits close behind, because it affects the majority of orders and is a single, well-scoped fix rather than a company-wide process change. Regional and category non-uniformity, by contrast, is best treated as an ongoing monitoring practice rather than a one-time fix, since the underlying pattern — strong aggregates hiding weak pockets — will keep recurring as the business grows.

5b.9 Conclusion

This project demonstrates how Power BI can improve supply chain visibility through interactive dashboards. Clean and well-structured data enables organizations to monitor performance, identify operational issues, and make faster data-driven decisions.

By following a structured workflow — from data collection and understanding, through connection and preprocessing, to dashboard creation and consolidation — the project delivers a reusable framework that can be extended with additional data sources, more advanced DAX measures, and further drill-through pages in future work. The insight-and-recommendation pairs developed in Section 5b.8 turn that framework from a descriptive report into a prescriptive one — each finding is tied to a specific, actionable next step rather than left as an observation alone.

6. Snapshots / Screenshots

This section presents the seven live dashboard pages built in Power BI Desktop, together with the data-preprocessing workspace that underpins them, exported directly from the working report file. Each screenshot is followed by a short business-intelligence theory note explaining what its KPIs and visuals mean, why that particular metric or chart type was chosen, and how it supports decision-making — tying the visual back to the objectives set out in Section 2.

6.1 Data Preprocessing – Power Query Editor

Before any visual could be built, the raw CSV export had to be cleaned inside Power Query. The screenshot below shows the “Retails Order Full Dataset” query open in the Power Query Editor. The Applied Steps pane on the right captures every transformation performed on the raw data — Promoted Headers, Changed Type, Removed Columns, Removed Blank Rows, Reordered Columns, Trimmed Text, Cleaned Text, and Sorted Rows — giving a fully auditable, repeatable data-preparation pipeline before the dataset is loaded into the Power BI data model.

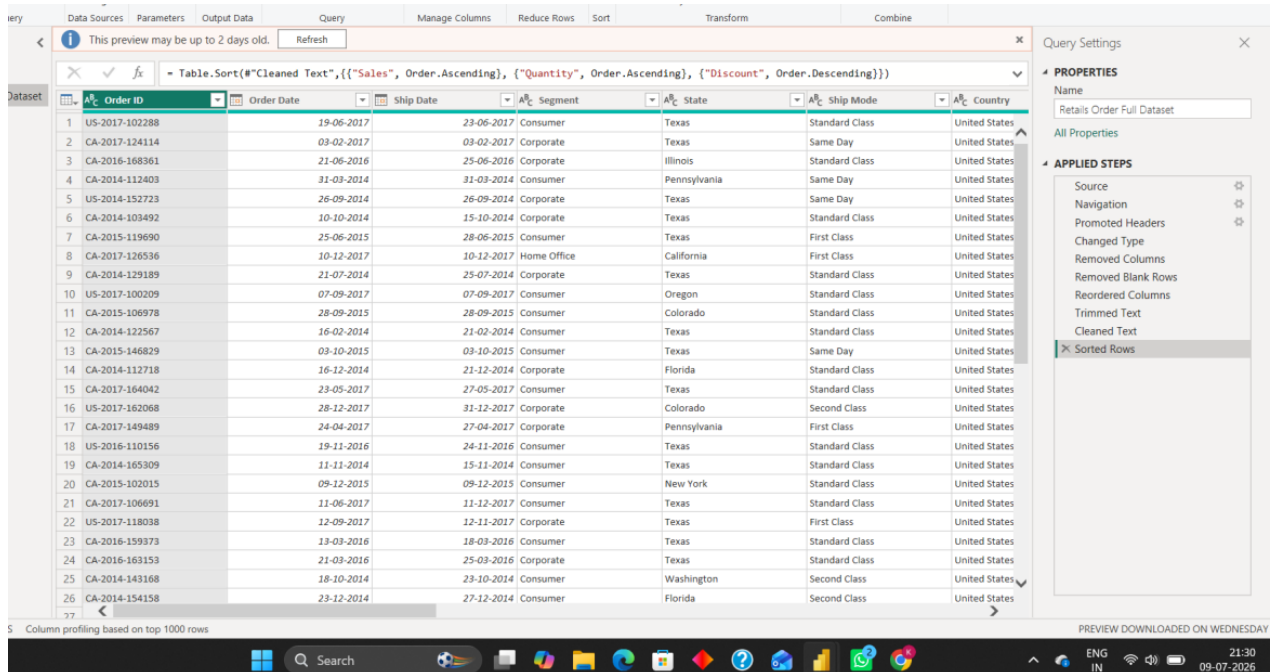


Figure 2: Power Query Editor — Retails Order Full Dataset with applied transformation steps.

Related Theory — Data Preprocessing

- **Promoted Headers & Changed Type:** Ensures every column is read with the correct data type (date, number, or text) rather than as generic text, which is a prerequisite for any DAX calculation to work correctly.
- **Removed Columns & Removed Blank Rows:** Strips out fields not needed for the intended visuals and eliminates blank records that would otherwise distort counts such as Total Orders.
- **Trimmed Text & Cleaned Text:** Removes stray whitespace and non-printing characters from categorical fields such as Segment, Region, and Ship Mode so that identical categories are not split into near-duplicate groups on bar and donut charts.
- **Sorted Rows:** Orders the query output for easier auditing during development; this step is cosmetic and does not affect the loaded data model.

6.2 Dashboard Page 1 – Executive Dashboard

Headline KPI cards for Total Sales, Total Profit, Total Orders, and Profit Margin, alongside Sales by Category, Profit by Region, and the Monthly Sales Trend.



Figure 3: Dashboard Page 1 — Executive Dashboard.

Related Theory — Executive KPIs

- **Total Sales (2.30M):** A summation measure representing gross revenue. As the top-line figure in any P&L structure, it indicates market demand and the overall scale of the business before any costs are subtracted.
- **Total Profit (286.41K):** The bottom-line measure after cost of goods and discounts. Distinguishing profit from sales prevents the common mistake of treating revenue growth as success on its own.
- **Total Orders (5,009):** A DISTINCTCOUNT of Order ID, representing operational throughput and a proxy for logistics and fulfillment workload.
- **Profit Margin (12.47%):** Total Profit ÷ Total Sales, expressed as a percentage — a normalized efficiency metric that allows fair comparison across time periods, regions, or categories.
- **Sales by Category (bar chart):** Ranks the three categories by revenue (Technology 0.84M, Furniture 0.74M, Office Supplies 0.72M) — a Pareto-style view informing inventory prioritization.
- **Profit by Region (bar chart):** A geographic performance comparison — West (108K) leads ahead of East (92K), South (47K), and Central (40K).
- **Monthly Sales Trend (line chart):** A time-series view spanning 2014–2017 that exposes seasonality, an overall upward growth trajectory, and month-over-month volatility.

6.3 Dashboard Page 2 – Supply Chain Insights

Return Rate and Returned Orders KPIs, alongside Sales by Segment, Total Quantity by Region, Top Products by Sales, Total Orders by Ship Mode, and Total Profit by Segment.



Figure 4: Dashboard Page 2 — Supply Chain Insights.

Related Theory — Supply Chain KPIs

- **Return Rate (5.91%):** Returned Orders ÷ Total Orders — a quality and customer-satisfaction proxy that drives reverse-logistics cost.
- **Returned Orders (296):** The absolute count behind the ratio above, used by operations teams to size reverse-logistics capacity.
- **Sales by Segment:** Consumer leads at 1.2M, ahead of Corporate (0.7M) and Home Office (0.4M), applying customer-segmentation theory.
- **Total Quantity by Region:** Maps physical demand intensity by geography — West (12.3K units) ahead of East, Central, and South.
- **Top Products by Sales:** An ABC/Pareto ranking of SKUs by revenue contribution, led by the Canon imageCLASS line at 62K.
- **Total Orders by Ship Mode (donut):** A modal-split view — Standard Class dominates at 59.77% of orders, illustrating the classic cost-versus-service trade-off.
- **Total Profit by Segment:** Segments profitability rather than just revenue — Consumer (134K), Corporate (92K), Home Office (60K).

6.4 Dashboard Page 3 – Product & Pricing Analysis

Average Discount and Average Shipping Days KPIs, alongside Top Products by Profit, Average Discount by Category, Sales by Sub-Category, Profit by Category, and Profit by Sub-Category.



Figure 5: Dashboard Page 3 — Product & Pricing Analysis.

Related Theory — Product & Pricing KPIs

- **Average Discount (0.09):** A key pricing lever measured as a decimal fraction of list price, used as a baseline for flagging unusual discounting.
- **Average Shipping Days (34.61):** A service-level KPI capturing lead time from order to shipment/delivery, tracked alongside cost metrics.
- **Top Products by Profit:** A product-level profitability ranking led by the Canon imageCLASS line at 25K.
- **Average Discount by Category:** Office Supplies is discounted most heavily (0.11), followed by Furniture (0.07) and Technology (0.01).
- **Sales by Sub-Category:** Phones and Chairs lead at 0.33M each, useful for merchandising decisions such as shelf space and bundling.
- **Profit by Category:** Technology (145K) and Office Supplies (122K) are strongly positive; Furniture (18K) is comparatively weak.
- **Profit by Sub-Category:** A ranked waterfall identifying Tables as the single biggest loss-maker at -18K, the most actionable view for pricing correction.

6.5 Dashboard Page 4 – Customer & Regional Insights

Total Quantity and Average Order Value KPIs, alongside Orders by Segment, Quantity by Category, Sales by State, Average Shipping Days, and Average Order Value by Segment.



Figure 6: Dashboard Page 4 — Customer & Regional Insights.

Related Theory — Customer & Regional KPIs

- **Total Quantity (38K):** A unit-volume KPI, distinct from revenue, that matters for warehouse throughput and shipping capacity planning.
- **Average Order Value (458.56):** Total Sales ÷ Total Orders — reveals whether upsell/cross-sell efforts are increasing what each customer spends.
- **Orders by Segment (donut):** Consumer accounts for 51.95% of orders, Corporate 30.22%, and Home Office 17.83%.
- **Quantity by Category:** Office Supplies leads at 23K units, giving a more balanced picture of demand than the value-based category view.
- **Sales by State:** California (0.46M) and New York (0.31M) anchor overall performance, applying the Pareto principle geographically.
- **Average Shipping Days:** Standard Class averages 42 days, Second Class 31, First Class 24, and Same Day just 1 day.
- **Average Order Value by Segment:** Home Office carries the highest average ticket (472), narrowly ahead of Corporate (466) and Consumer (449).

6.6 Dashboard Page 5 – Business Performance Monitoring

Distinct Products and States Served KPIs, alongside Loss by Product Name, Sum of Profit by State, Sum of Profit by Returned, Count of Order ID by Returned, and Average of Discount by Ship Mode.



Figure 7: Dashboard Page 5 — Business Performance Monitoring.

Related Theory — Business Performance KPIs

- **Distinct Products (2K):** A catalogue-breadth KPI reflecting the width of the product assortment the business carries.
- **States Served (49):** A market-coverage KPI indicating the geographic reach of the business.
- **Loss by Product Name:** Ranks the worst-performing products directly by loss — the Cubify CubeX 3D Printer alone accounts for -8.9K.
- **Sum of Profit by State:** A geographic profitability view led by California (76K) and New York (74K).
- **Sum of Profit by Returned:** Non-returned orders carry 0.26M in profit versus only 0.02M for returned orders.
- **Count of Order ID by Returned:** 9.19K orders (91.99%) not returned against 0.8K (8.01%) returned — a quality-control cross-check.
- **Average of Discount by Ship Mode:** First Class and Standard Class both average 0.09, Same Day 0.08, and Second Class the lowest at 0.07.

6.7 Dashboard Page 6 – Discount & Profitability Diagnostics

Delivered in Milestone-4: Profitable Order Rate, Loss-Making Order Rate, and Discounted Order Rate KPIs, alongside Average Discount / Sum of Profit / Total Orders by Category, Total Profit by Discount Band, Total Profit by Profitability Band, and Total Profit by Category and Segment.

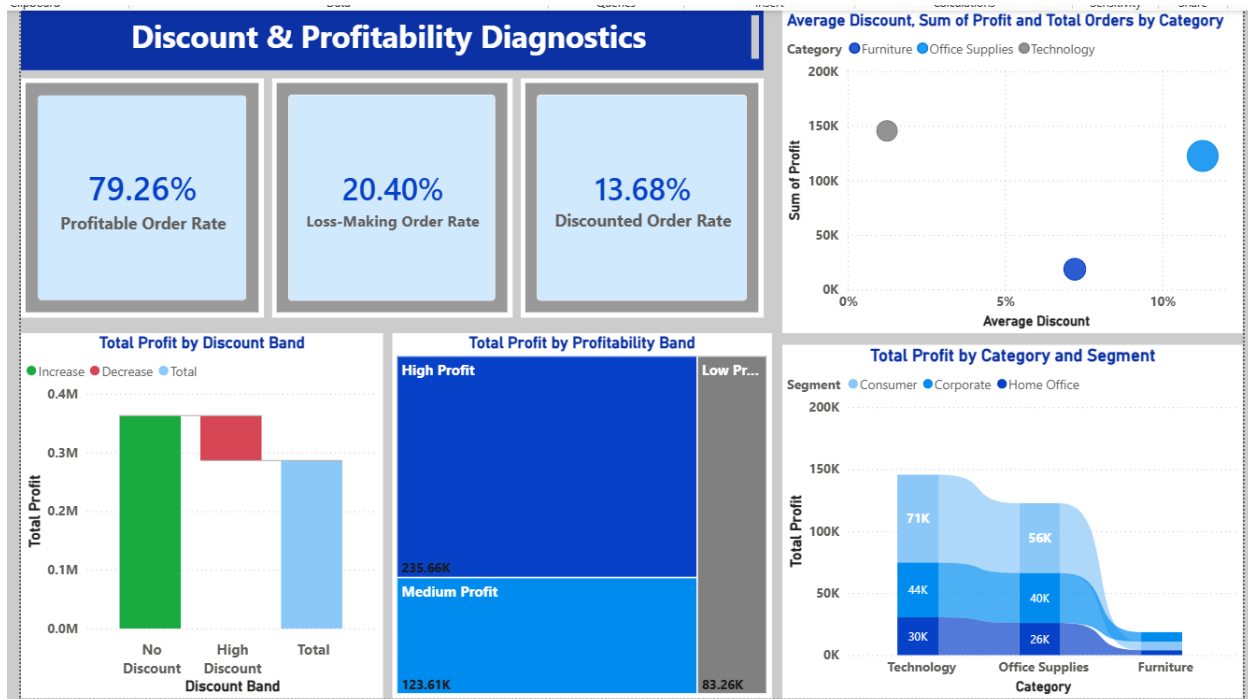


Figure 8: Dashboard Page 6 — Discount & Profitability Diagnostics.

Related Theory — Discount & Profitability KPIs

- **Profitable Order Rate (79.26%):** The share of orders that close with a positive profit — the primary health check for the order book before discount policy is examined.
- **Loss-Making Order Rate (20.40%):** One order in five currently loses money, making this the single most actionable KPI on the page for tightening pricing and discount rules.
- **Discounted Order Rate (13.68%):** The proportion of orders that carry any discount at all, giving context for how much of the loss-making rate above discounting can plausibly explain.
- **Average Discount, Sum of Profit and Total Orders by Category (scatter/bubble):** Plots each category's average discount against its total profit, with bubble size for order volume — Technology sits in the low-discount, high-profit corner while Furniture sits in the higher-discount, lower-profit corner.
- **Total Profit by Discount Band (waterfall):** Shows profit stepping down from the No-Discount band through a High-Discount decrease to the Total, quantifying exactly how much profit discounting removes from the business.
- **Total Profit by Profitability Band (treemap):** Segments all profit into High Profit (235.66K), Medium Profit (123.61K), and Low Profit (83.26K) bands, showing where the bulk of profit is concentrated.
- **Total Profit by Category and Segment (ribbon chart):** Breaks profit down by both category and customer segment simultaneously — Technology–Consumer (71K) and Office Supplies–Consumer (56K) are the two largest contributing combinations.

6.8 Dashboard Page 7 – Order & Unit Economics

Delivered in Milestone-4: Average Items per Order, Sales per Unit, and Profit per Unit KPIs, alongside Profit per Unit by Category and Segment, Profit per Unit by Category, Profit per Unit by Sub-Category, and Average Items per Order by Segment.

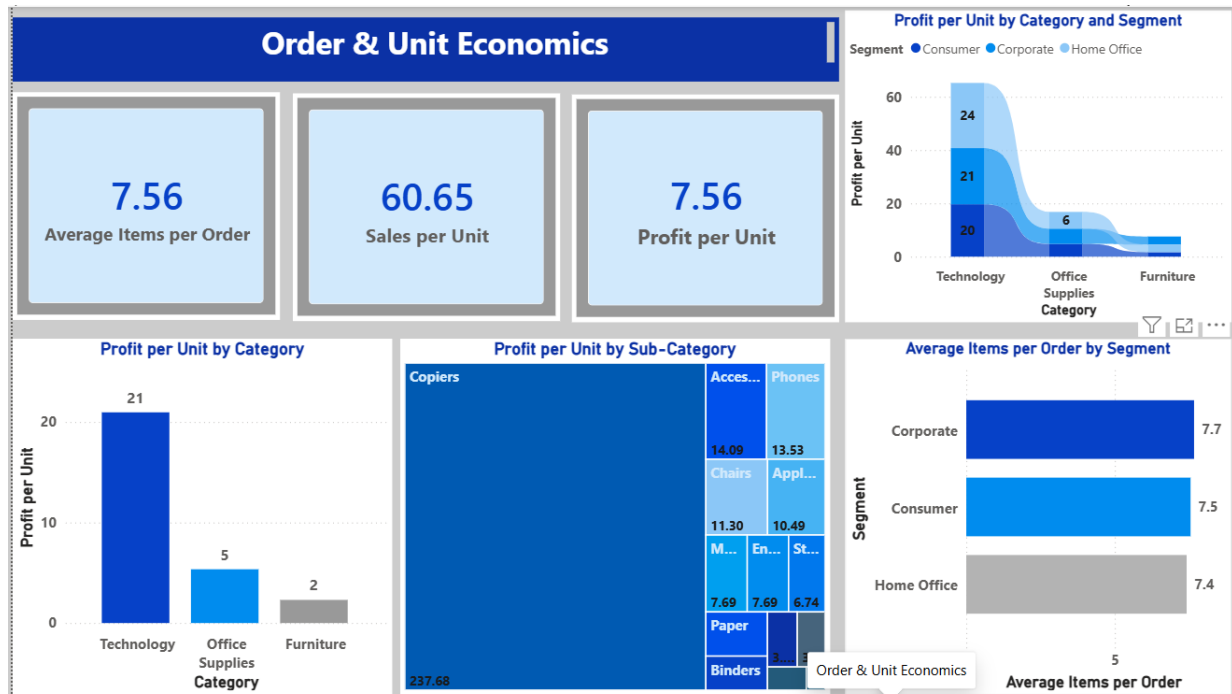


Figure 9: Dashboard Page 7 — Order & Unit Economics.

Related Theory — Order & Unit Economics KPIs

- **Average Items per Order (7.56):** A basket-size KPI showing how many line items a typical order contains, independent of order value.
- **Sales per Unit (60.65):** Normalizes revenue by unit sold rather than by order, giving a cleaner comparison of pricing across categories with very different basket sizes.
- **Profit per Unit (7.56):** The unit-level profitability counterpart to Sales per Unit, used together to see how much of each unit's price actually converts to profit.
- **Profit per Unit by Category and Segment (ribbon chart):** Shows Technology consistently generating the highest per-unit profit across all three customer segments, with Office Supplies and Furniture well behind.
- **Profit per Unit by Category (bar chart):** Technology leads sharply at 21 per unit, against 5 for Office Supplies and just 2 for Furniture — confirming Technology as the highest-value category per item sold, not only in aggregate.
- **Profit per Unit by Sub-Category (treemap):** Copiers dominate at 237.68 per unit, an order of magnitude above every other sub-category such as Accessories (14.09), Phones (13.53), and Chairs (11.30) — the clearest single candidate for continued investment and stock priority.
- **Average Items per Order by Segment (bar chart):** Corporate (7.7), Consumer (7.5), and Home Office (7.4) order broadly similar basket sizes, showing that segment differences in revenue come from price and mix rather than basket size.

7. Challenges Faced

A number of technical, operational, and communication challenges came up over the course of the project. Each is described below along with how it was resolved or mitigated.

- **Inconsistent categorical values:** Region, Segment, and Ship Mode values in the raw CSV occasionally differed only in casing or spacing, which risked splitting a single category into near-duplicate groups on bar and donut charts. This was resolved during the Data Preprocessing stage by standardizing text formatting for every categorical column before the data was loaded into the model.
- **Relationship design in the data model:** Deciding whether to keep order date and ship date on the same relationship or separate them required care, since an incorrect relationship would silently produce wrong totals on time-based visuals. This was resolved by adding a single active relationship on order date and documenting how ship date could be activated separately using DAX (USERRELATIONSHIP) if needed.
- **Keeping new pages consistent with the existing design:** Adding five additional dashboard pages across Milestone-3 and Milestone-4 without the layout, colours, or measure logic drifting from Pages 1 and 2 required deliberate reuse of the existing measure library and visual formatting rather than starting each new page from a blank canvas.
- **Coordinating work across a six-person team within a compressed eight-week schedule:** This was mitigated by dividing work along the same phase boundaries as the milestones themselves, and by using Microsoft Teams for stand-ups and GitHub/GitLab for version control, so that data preparation, dashboard design, and documentation could proceed with clear handoff points rather than everyone working on the same file at once.

8. Learnings & Skills Acquired

8.1 Technical Skills

Hands-on proficiency was developed across the core Power BI toolchain: connecting and shaping data in Power Query, designing a simple relational data model with fact and dimension tables, and writing DAX measures that go beyond simple sums to include ratios such as Profit Margin and Return Rate, as well as time-intelligence measures for year-over-year growth. Beyond the measure library itself, the project built a working understanding of when to reach for each visual type — card visuals for a single headline KPI, bar charts for ranked category comparison, a donut for part-to-whole share, a waterfall for a stepped change such as a discount-band profit drop, a treemap for a hierarchical breakdown such as profit by sub-category, and a scatter or bubble chart for exploring the relationship between two continuous measures such as discount depth and profit. Choosing the wrong visual type for a given question was a recurring early mistake — for example, an early draft used a bar chart where a waterfall later proved far clearer for showing how profit is eroded step by step — and correcting it became a useful, repeatable lesson: the shape of the business question should decide the chart type, not the other way around. Working across seven interlinked report pages also built real comfort with Power BI's cross-filtering and cross-report drill-through model, and with the discipline of testing every slicer and filter combination rather than assuming a working visual on one page implies a working interaction on every page.

8.2 Analytical and Problem-Solving Skills

Working through real transactional data built a habit of asking what a number actually means before charting it — for example, recognizing that a raw discount column only becomes useful once compared directly against profit, which is what motivated the discount-versus-profit scatter plot on the dashboard. This habit extended into how the team read its own results: a strong regional or category total was never accepted at face value without checking whether it held up at the next level of granularity, which is exactly the discipline that surfaced the finding that West-region performance is not uniform across every state. The team also learned to be skeptical of a single KPI in isolation — the Loss-Making Order Rate only became genuinely useful once compared against the Discounted Order Rate, because the gap between the two numbers is what revealed that discounting alone does not explain every loss-making order. This kind of comparative, cross-checking analysis — reading two or three related KPIs together rather than each one on its own — became the team's default way of validating an insight before writing it down as a recommendation.

8.3 Future Enhancements

Several extensions were identified for future work beyond this internship: introducing a scheduled data refresh so the dashboard stays current automatically rather than relying on a static CSV snapshot; exploring row-level security so different regional managers could see a filtered view of the same report without needing separate files; extending the loss-making-order analysis into the discount-driven versus structurally-thin-margin split identified in Section 5b.8; adding a drill-through path from category to sub-category and from region to state directly on the Executive Dashboard; and building a small what-if parameter for discount depth so a manager could simulate the profit impact of a proposed discount policy change before applying it. None of these were in scope for Milestone-4, but each follows directly from a gap the team noticed while building and testing the current seven-page report, which is itself a useful outcome — the project ended with a concrete, prioritized backlog rather than a vague sense that "more could be done."

8.4 Soft Skills and Team Collaboration

Regular check-ins with the mentor and the rest of Team B — coordinated through Microsoft Teams — reinforced the value of agreeing on scope and terminology early, such as exactly how "Return Rate" should be defined, so that later work did not need to be redone. Version control through GitHub, with GitLab as a backup repository, made it easier to review each other's work before it was merged into the shared report file. Splitting the six-person team's work along the same phase boundaries as the milestones themselves — rather than by dashboard page — turned out to be an important decision: it meant data preparation, dashboard design, and documentation could proceed with clear handoff points, and it reduced the number of times two people were editing the same Power BI file at once, which had been an early source of merge conflicts. Presenting interim progress to the mentor at the midterm review also proved valuable beyond the feedback itself — having to explain the dashboard's design choices out loud surfaced a few visuals that looked fine individually but did not clearly answer a business question, which the team then reworked before they became embedded habits carried into later pages.

8.5 Domain Knowledge and Application

Beyond the tooling, the project built a working understanding of the supply chain domain itself — how orders flow from placement to shipment to delivery, why metrics like lead time and on-time delivery rate matter to a business, and how discounting decisions interact with profitability at a product level. Reading the Retail Supply Chain Sales Dataset field by field in Milestone-1, well before any chart was built, turned out to be time well spent: understanding what "ship mode" and "order priority" actually represented in a retail operation made it much easier, later, to judge whether a given visual was asking a sensible business question or simply displaying a column because the data happened to be available. That same domain grounding is what allowed the team to interpret the Milestone-4 additions — profitable-order rate, loss-making-order rate, and unit economics — as extensions of a coherent supply-chain story rather than as a disconnected set of new numbers bolted onto the end of the report.

8.6 Documentation and Communication Skills

A further, less expected learning was around documentation discipline itself. Writing this Completion Report alongside the dashboard — rather than after it was finished — made it much easier to capture the reasoning behind a design decision while it was still fresh, instead of trying to reconstruct it weeks later. It also surfaced a few inconsistencies between what the team intended a measure to do and what it actually calculated, which were caught and corrected earlier than they would have been otherwise. Presenting technical work to a non-technical audience — writing the KPI theory notes in Section 6 so that a reader unfamiliar with Power BI could still follow what each visual meant and why it mattered — was itself a useful exercise in translating analytical work into plain business language, a skill that carries over directly into any future data or software role.

9. Testimonials from Team

P. Venkata Bharath

"Working on the Supply Chain Visibility & Optimization Dashboard over these eight weeks has been one of the most practical, skill-building experiences of my academic journey so far. Starting from a raw CSV file and ending with a seven-page, fully interactive Power BI report made every stage of the analytics process real to me — not just something I had read about. Building the DAX measures and watching every card and chart update instantly as I applied a filter was genuinely satisfying, and it taught me to think about data quality and modeling decisions far more carefully than I used to. I am grateful to Ms. Nityasree and the Infosys Springboard programme for the structure and guidance that made this possible, and to my teammates for making the collaboration smooth throughout."

10. Conclusion

This project set out to improve supply chain visibility for a retail business using Power BI, and by the end of the internship that objective was met through a complete, seven-page interactive dashboard built on a clean, well-modeled dataset. Milestone-1 established a solid understanding of the data and the business problem; Milestone-2 turned that understanding into a working two-page dashboard with a reusable DAX measure library; Milestone-3 extended that foundation into a fuller analytical suite covering regional, category, and logistics performance; and Milestone-4 consolidated, validated, and finalized the complete seven-page dashboard — adding discount and profitability diagnostics and order and unit economics — for stakeholder presentation and handover. Each milestone was designed to build directly on the one before it rather than starting fresh, which is why the same data model, the same nine-measure DAX library, and the same visual language carry through from Page 1 of the Executive Dashboard all the way to Page 7 of Order & Unit Economics without drifting in formatting, colour, or logic.

Throughout the project, the emphasis stayed on producing a dashboard that a real supply chain or sales manager could use to make faster, better-informed decisions — not simply a set of charts arranged on a page. That distinction shaped almost every design choice along the way: a visual was only added if it answered a specific business question a manager would plausibly ask, and every KPI card was chosen because it normalizes, ranks, or contextualizes the underlying numbers rather than simply restating a raw column total. The insights drawn out along the way demonstrate that this approach paid off in practice. The strength of the West region, once isolated on the Executive Dashboard, led directly to the recommendation to study and pilot its operating practices elsewhere. The impact of heavy discounting on profit, first visible as a loose pattern on the Product & Pricing Analysis page in Milestone-3, was sharpened into a precise, quantified finding once the Discount & Profitability Diagnostics page was added in Milestone-4 — the business now has an exact figure, 20.40%, for how many orders are currently loss-making, and a clear hypothesis, discount depth in Furniture and Office Supplies, for where to start fixing it. The unit-economics view on the final page went a step further, showing that Technology's advantage holds even after controlling for order size, with Copiers standing out as a genuinely exceptional sub-category worth protecting from stock-outs.

The technical work behind these insights was substantial and is documented in detail across Sections 5 and 6 of this report, but it is worth restating briefly here because it is what makes the insights trustworthy rather than merely plausible. Every KPI on every page traces back to the same small library of DAX measures, defined once in Milestone-2 and reused without modification across all seven pages — Total Sales, Total Profit, Total Orders, Total Quantity, Profit Margin, Return Rate, Returned Orders, Average Profit, and Average Discount. Centralizing these calculations rather than rebuilding similar logic on each page meant that a fix or refinement made in one place propagated correctly everywhere, and it meant every number in this report can be traced back to a single, auditable definition rather than several slightly

different ad-hoc formulas. The data-preparation work in Power Query — removing duplicate order and shipment records, handling missing shipping and delivery dates, standardizing categorical fields such as Segment, Region, and Ship Mode — is easy to overlook once the finished dashboard is in front of a stakeholder, but it is precisely what makes every downstream number reliable; a return-rate figure built on top of duplicated records, for instance, would have been meaningfully wrong from the very first milestone.

Working as a six-person team within a compressed eight-week schedule was its own test of process discipline, separate from the analytical work itself. Dividing responsibilities along the same phase boundaries as the milestones — rather than by dashboard page — kept data preparation, dashboard design, and documentation moving with clear handoff points, and using GitHub as the primary version-control repository with GitLab as a backup gave the team a recoverable history of the project at every stage. Regular mentor check-ins through Microsoft Teams were particularly valuable at the transition points between milestones, where the team had to decide how much of the existing dashboard to keep unchanged versus how much to expand — the decision, ultimately, to always extend the existing measure library and visual language rather than rebuild it from scratch is the reason the finished seven-page report reads as one coherent product rather than four report fragments stitched together.

Overall, this internship has been a meaningful, hands-on introduction to professional business intelligence work, and it aligns closely with the academic and career direction of the team members involved. Moving from a raw CSV export to a governed, interactive, seven-page Power BI report — with a documented data model, a reusable DAX measure library, and a written record of every design decision along the way — mirrors, in miniature, the kind of work a business intelligence or data analytics team does in industry, and it is exactly the kind of applied, end-to-end experience that is difficult to replicate in a purely academic setting. The skills, habits, and domain knowledge built here — from data modeling discipline and DAX measure design, to reading a KPI in the context of its neighbours rather than in isolation, to clear business communication and collaborative version control — are expected to carry forward well beyond this project, both into further coursework and into future internship and placement opportunities in data analytics, business intelligence, and software engineering roles.

Looking ahead, the backlog of extensions identified in Section 8.3 — scheduled data refresh, row-level security for regional managers, a discount-depth what-if parameter, and deeper drill-through paths between category and sub-category and between region and state — gives a clear, prioritized direction for any future iteration of this dashboard, whether that iteration happens as a later internship milestone or as an independent follow-on project. None of these extensions require rebuilding the current work; each one builds on the same data model and measure library that has carried the report from its first two pages through to its complete seven-page form, which is itself a small but genuine validation of the modeling decisions made back in Milestone-1. In that sense, the project closes not with a finished, static artifact, but with a working, extensible foundation — which was the actual goal from the very first milestone.

Reflecting on the eight weeks as a whole, the most durable outcome of this internship is not any single chart or KPI, but the repeatable process behind them: understand the business problem before opening the tool, clean and model the data before building a single visual, define every calculation once as a governed measure rather than several inconsistent ad-hoc ones, and validate every insight against a second, related KPI before writing it down as a recommendation. That process is what allowed a six-person team to take a raw CSV export from Kaggle and turn it, across four connected milestones, into a seven-page, cross-filtered, stakeholder-ready Power BI report without the analytical thread ever breaking along the way. It is also what makes the eight insight-and-recommendation pairs in Section 5b.8 more than a list of interesting numbers — each one is traceable back through a specific visual, a specific measure, and a specific preprocessing decision, all the way to the original field definitions captured in Milestone-1. Closing this final Completion Report covering all four milestones, the team is confident that the dashboard, the documentation, and the process behind both would transfer cleanly to a real retail supply-chain team looking to replace fragmented, ad-hoc reporting with a single, governed source of truth.

11. Acknowledgements

We would like to express our sincere gratitude to the Infosys Springboard Team for organizing this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

We extend our heartfelt appreciation to our mentor, Ms. Nityasree, for her consistent guidance, valuable feedback, and continuous encouragement throughout the project lifecycle. Her insights and structured direction played a significant role in shaping the analytical depth and professional quality of this project.

We would also like to acknowledge our team members for their strong collaboration, accountability, and shared commitment toward achieving project objectives. The coordinated distribution of responsibilities, open communication through Microsoft Teams, and collective problem-solving approach significantly contributed to the successful execution of this project.

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The structured learning resources and practical exposure provided through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. This experience has enhanced our ability to approach data with a strategic mindset, collaborate effectively within a team environment, and execute projects with professional discipline.

Overall, this internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.